```
/* نماذج السيارات (كامري 2021 + هايلكس 2016) — مضغوطة */
const CAR_META={"camry":{"body":{"paint":{"nv":6636,"ni":35358,"vo":0,"io":39816,
const CAR_BIN="DzLl7Zl5XzJI75V8MDLl7Zl5gTJI75V8TzLv7Zl51zFA7NZ2tTFA7NZ29jFK7NV2CD
'use strict';
/* ============================================================
   ملاحة حائل 3D — app.js
   خريطة + سيارة ثلاثية الأبعاد + عداد سرعة + تعليمات صوتية
   + كاميرا ذكاء اصطناعي لكشف السيارات والتنبيه من الحوادث
   ============================================================ */

const HAIL = [41.6907, 27.5219];                    // وسط مدينة حائل
const BOX = { w: 38.3, s: 25.3, e: 44.4, n: 29.4 };   // حدود منطقة حائل للبحث
const STYLE = 'https://tiles.openfreemap.org/styles/dark';
const OSRM = 'https://router.project-osrm.org/route/v1/driving/';
const NOM = 'https://nominatim.openstreetmap.org/';
const TFJS = 'https://cdn.jsdelivr.net/npm/@tensorflow/tfjs@4.22.0/dist/tf.min.js
const COCO = 'https://cdn.jsdelivr.net/npm/@tensorflow-models/coco-ssd@2.2.3/dist
const NAV_PITCH = 70, NAV_ZOOM = 18.2;

const $ = id => document.getElementById(id);
const clamp = (v, a, b) => Math.max(a, Math.min(b, v));

const st = {
  mode: 'plan', gps: null, lastG: null, dest: null, route: null,
  coords: [], cum: [], steps: [], maxs: [],
  fixAlong: 0, fixT: 0, dispAlong: 0, speed: 0, kmh: 0, accel: 0, idx: 0,
  onRoute: true, offCount: 0, rawPos: null, lastRaw: null, rawHead: 0, dispHead:
  sim: false, simTimer: null, free: false, freeTimer: null, muted: false,
  lastReroute: 0, rerouting: false, ann: {}, arrived: false, limit: null, lastOve
  wake: null, lastLineUpd: 0, navPad: { top: 0, bottom: 0, left: 0, right: 0 }
};

/* ---------------- سيارتك ---------------- */
const CAR_COLORS = [0xf2f2f0, 0xbfc3c8, 0x17181b, 0x6d7279, 0x8b1a1a, 0x1f3f7a, 0
const myCar = (() => {
  let v = { type: 'camry', color: 0xf2f2f0 };
  try { const j = JSON.parse(localStorage.getItem('hailnav_car') || 'null'); if (
  return v;
})();
function saveCar() { try { localStorage.setItem('hailnav_car', JSON.stringify(myC

/* ---------------- أدوات جغرافية ---------------- */
const RAD = Math.PI / 180;
```

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function dist(a, b) {
  const dLat = (b[1] - a[1]) * RAD, dLng = (b[0] - a[0]) * RAD;
  const x = Math.sin(dLat / 2) ** 2 + Math.cos(a[1] * RAD) * Math.cos(b[1] * RAD)
  return 2 * 6371000 * Math.asin(Math.sqrt(x));
}
function bearing(a, b) {
  const y = Math.sin((b[0] - a[0]) * RAD) * Math.cos(b[1] * RAD);
  const x = Math.cos(a[1] * RAD) * Math.sin(b[1] * RAD) - Math.sin(a[1] * RAD) *
  return (Math.atan2(y, x) / RAD + 360) % 360;
}
const angDiff = (a, b) => ((b - a + 540) % 360) - 180;
function segAt(al) {
  const cum = st.cum; let lo = 0, hi = cum.length - 1;
  if (al <= 0) return 0; if (al >= cum[hi]) return hi - 1;
  while (hi - lo > 1) { const m = (lo + hi) >> 1; if (cum[m] <= al) lo = m; else
  return lo;
}
function pointAt(al) {
  const c = st.coords, cum = st.cum;
  if (!c.length) return null;
  if (al <= 0) return c[0];
  if (al >= cum[cum.length - 1]) return c[c.length - 1];
  const i = segAt(al), t = (al - cum[i]) / ((cum[i + 1] - cum[i]) || 1);
  return [c[i][0] + (c[i + 1][0] - c[i][0]) * t, c[i][1] + (c[i + 1][1] - c[i][1]
}
function snap(p) {
  const c = st.coords; if (c.length < 2) return null;
  const kx = Math.cos(p[1] * RAD) * 111320, ky = 110540;
  let best = null;
  const scan = (from, to) => {
    for (let i = from; i <= to; i++) {
      const ax = (c[i][0] - p[0]) * kx, ay = (c[i][1] - p[1]) * ky;
      const bx = (c[i + 1][0] - p[0]) * kx, by = (c[i + 1][1] - p[1]) * ky;
      const vx = bx - ax, vy = by - ay, L2 = vx * vx + vy * vy || 1e-9;
      const t = clamp(-(ax * vx + ay * vy) / L2, 0, 1);
      const px = ax + vx * t, py = ay + vy * t, d = Math.hypot(px, py);
      if (!best || d < best.d) best = { d, i, along: st.cum[i] + (st.cum[i + 1] -
    }
  };
  scan(Math.max(0, st.idx - 30), Math.min(c.length - 2, st.idx + 300));
  if (!best || best.d > 60) scan(0, c.length - 2);
  return best;
}
function fmtDist(m) {
  if (m < 1000) return `${Math.max(10, Math.round(m / 10) * 10)} م`;
  return `${(m / 1000).toFixed(m < 10000 ? 1 : 0)} كم`;
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}
function fmtTime(sec) {
  const m = Math.round(sec / 60);
  if (m < 60) return `${Math.max(1, m)} د`;
  return `${Math.floor(m / 60)} س ${m % 60} د`;
}
function clock(sec) {
  const d = new Date(Date.now() + sec * 1000);
  let h = d.getHours(); const ap = h >= 12 ? 'ص' : 'م'; h = h % 12 || 12;
  return `${h}:${String(d.getMinutes()).padStart(2, '0')} ${ap}`;
}

/* ---------------- رسائل وصوت ---------------- */
let toastT;
function toast(msg, ms = 3200) {
  const t = $('toast'); t.textContent = msg; t.classList.add('on');
  clearTimeout(toastT); toastT = setTimeout(() => t.classList.remove('on'), ms);
}
let arVoice = null;
function pickVoice() {
  if (!('speechSynthesis' in window)) return;
  const v = speechSynthesis.getVoices();
  // نفضّل صوت رجالي عربي إذا متوفر
  arVoice = v.find(x => /^ar/i.test(x.lang) && /maged|tarik|hamed|male|رجل/i.test
            v.find(x => /^ar[-_]SA/i.test(x.lang)) || v.find(x => /^ar/i.test(x.l
}
if ('speechSynthesis' in window) { pickVoice(); speechSynthesis.onvoiceschanged =
const lastSaid = {};
function speak(text, urgent = false, key = text, gap = 8000) {
  if (st.muted || !('speechSynthesis' in window)) return;
  const now = Date.now(); if (lastSaid[key] && now - lastSaid[key] < gap) return;
  lastSaid[key] = now;
  if (urgent) speechSynthesis.cancel();
  const u = new SpeechSynthesisUtterance(text);
  u.lang = 'ar-SA'; if (arVoice) u.voice = arVoice; u.rate = 1.02;
  speechSynthesis.speak(u);
}
let actx = null;
function beep(freq = 880, dur = 0.12, vol = 0.25) {
  if (st.muted) return;
  try {
    actx = actx || new (window.AudioContext || window.webkitAudioContext)();
    const o = actx.createOscillator(), g = actx.createGain();
    o.type = 'square'; o.frequency.value = freq; g.gain.value = vol;
    o.connect(g); g.connect(actx.destination); o.start();
    g.gain.exponentialRampToValueAtTime(0.0001, actx.currentTime + dur); o.stop(a
```

```
  } catch (e) {}
}
const alertCool = {};
let alertT;
function showAlert(key, text, sub = '', icon = '⚠️', level = 'warn', ms = 3500, c
  const now = Date.now();
  if (alertCool[key] && now - alertCool[key] < cool) return false;
  alertCool[key] = now;
  const el = $('alert');
  el.className = level; $('aIcon').textContent = icon; $('aText').textContent = t
  void el.offsetWidth; el.style.animation = 'none'; void el.offsetWidth; el.style
  clearTimeout(alertT); alertT = setTimeout(() => el.classList.add('hidden'), ms)
  return true;
}

/* ---------------- عداد السرعة ---------------- */
const G = { cx: 160, cy: 160, r: 128, a0: 168, a1: 12, max: 200 };
function gp(a, r = G.r) { return [G.cx + r * Math.cos(a * RAD), G.cy - r * Math.s
function valAng(v) { return G.a0 - (G.a0 - G.a1) * clamp(v / G.max, 0, 1); }
let gaugeArc, gaugeLen = 1, limitMark;
function buildGauge() {
  const svg = $('gauge'), [x0, y0] = gp(G.a0), [x1, y1] = gp(G.a1);
  const arc = `M${x0} ${y0} A${G.r} ${G.r} 0 0 1 ${x1} ${y1}`;
  let h = `<defs><linearGradient id="gg" x1="0" x2="1"><stop offset="0" stop-colo
  <filter id="gl"><feGaussianBlur stdDeviation="2.5"/><feMerge><feMergeNode/><feM
  <path d="${arc}" fill="none" stroke="rgba(255,255,255,.12)" stroke-width="7" st
  <path id="gArc" d="${arc}" fill="none" stroke="url(#gg)" stroke-width="7" strok
  for (let v = 0; v <= G.max; v += 10) {
    const a = valAng(v), big = v % 40 === 0;
    const [ax, ay] = gp(a, G.r - 12), [bx, by] = gp(a, G.r - (big ? 22 : 17));
    h += `<line x1="${ax}" y1="${ay}" x2="${bx}" y2="${by}" stroke="rgba(255,255,
    if (big) { const [tx, ty] = gp(a, G.r - 36); h += `<text x="${tx}" y="${ty +
  }
  h += `<path id="gLim" d="M0 -7 L6 5 L-6 5 Z" fill="#ffd36b" opacity="0"/>`;
  svg.innerHTML = h;
  gaugeArc = $('gArc'); gaugeLen = gaugeArc.getTotalLength();
  gaugeArc.style.strokeDasharray = gaugeLen; gaugeArc.style.strokeDashoffset = ga
  limitMark = $('gLim');
}
function setGauge(kmh) {
  $('spdNum').textContent = Math.round(kmh);
  gaugeArc.style.strokeDashoffset = gaugeLen * (1 - clamp(kmh / G.max, 0, 1));
}
function setLimit(lim) {
  const el = $('limit');
  if (!lim) { el.classList.add('unk'); el.firstChild.textContent = '--'; limitMar
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```
  el.classList.remove('unk'); el.firstChild.textContent = lim;
  const a = valAng(lim), [x, y] = gp(a, G.r + 11);
  limitMark.setAttribute('transform', `translate(${x} ${y}) rotate(${90 - a + 180
  limitMark.setAttribute('opacity', 1);
}

/* ---------- أسهم الاتجاهات ---------- */
const MOD_ANG = { straight: 0, 'slight right': 40, right: 90, 'sharp right': 135,
function arrowSVG(type, mod, color = '#ffd36b') {
  const s = `stroke="${color}" stroke-width="4.5" fill="none" stroke-linecap="rou
  if (type === 'arrive') return `<svg viewBox="0 0 48 48"><path d="M14 44V6 M14 8
  if (mod === 'uturn') return `<svg viewBox="0 0 48 48"><path d="M30 44V20a8 8 0
  if (type === 'roundabout' || type === 'rotary') return `<svg viewBox="0 0 48 48
  const a = MOD_ANG[mod] ?? 0;
  return `<svg viewBox="0 0 48 48"><path d="M24 44V26" ${s}/><g transform="rotate
}
const ORD = ['', 'السابع', 'السادس', 'الخامس', 'الرابع', 'الثالث', 'الثاني', 'لأول
function stepText(s) {
  const t = s.maneuver.type, m = s.maneuver.modifier || '', name = s.name || s.re
  const to = name ? ` إلى ${name}` : '';
  const dir = { left: 'يسار', 'sharp left': 'يسار بحدّة', 'slight left': 'سار خفيف
  if (t === 'arrive') return 'وصلت إلى وجهتك';
  if (t === 'depart') return `انطلق${to}`;
  if (t === 'roundabout' || t === 'rotary' || t === 'exit roundabout' || t === 'e
    return `ادخل الدوّار واطلع من المخرج ${ORD[s.maneuver.exit] || s.maneuver.exit
  if (m === 'uturn') return `لفّ وارجع (يو تيرن)${to}`;
  if (t === 'fork') return `خذ ${m.includes('left') ? 'اليمين' : 'اليسار'} التفرّع
  if (t === 'merge') return `اندمج ${m.includes('left') ? 'يمين' : 'يسار'}${to}`;
  if (t === 'on ramp') return `خذ المدخل ${m.includes('left') ? 'اليمين' : 'ليسار
  if (t === 'off ramp') return `اطلع من المخرج ${m.includes('left') ? 'اليمين' :
  if (t === 'continue' || t === 'new name' || m === 'straight' || !dir) return `م
  return `لف ${dir}${to}`;
}
function sideOf(s) {
  const m = s.maneuver.modifier || '', t = s.maneuver.type;
  if (t === 'arrive' || m === 'straight' || !m) return 0;
  if ((t === 'continue' || t === 'new name') && m.startsWith('slight')) return 0;
  return m.includes('left') || m === 'uturn' ? -1 : m.includes('right') ? 1 : 0;
}

/* ---------- الخريطة ---------- */
const map = new maplibregl.Map({
  container: 'map', style: STYLE, center: HAIL, zoom: 12.3, maxPitch: 85,
  attributionControl: { compact: true }, fadeDuration: 0
});
let meMarker = null, destMarker = null;
```

```
map.on('load', () => {
  const style = map.getStyle();
  // أسماء عربية
  for (const l of style.layers) {
    if (l.type !== 'symbol' || !l.layout || !l.layout['text-field']) continue;
    if (/shield|housenum|ref/i.test(l.id)) continue;
    try { map.setLayoutProperty(l.id, 'text-field', ['coalesce', ['get', 'name:ar
  }
  // مباني ثلاثية الأبعاد
  const src = Object.keys(style.sources).find(k => style.sources[k].type === 'vec
  const firstSym = style.layers.find(l => l.type === 'symbol');
  if (src) {
    try {
      map.addLayer({
        id: 'b3d', type: 'fill-extrusion', source: src, 'source-layer': 'building
        paint: {
          'fill-extrusion-color': ['interpolate', ['linear'], ['coalesce', ['get'
          'fill-extrusion-height': ['coalesce', ['get', 'render_height'], 6],
          'fill-extrusion-base': ['coalesce', ['get', 'render_min_height'], 0],
          'fill-extrusion-opacity': 0.92
        }
      }, firstSym && firstSym.id);
    } catch (e) {}
  }
  try { map.setSky({ 'sky-color': '#0a1326', 'horizon-color': '#26324d', 'fog-col
  // خط المسار الأصفر
  map.addSource('route', { type: 'geojson', data: emptyLine() });
  const rw = ['interpolate', ['exponential', 1.6], ['zoom'], 10, 4, 14, 8, 17, 22
  map.addLayer({ id: 'route-glow', type: 'line', source: 'route', layout: { 'line
    paint: { 'line-color': '#ffb21a', 'line-width': rw, 'line-blur': 10, 'line-op
  map.addLayer({ id: 'route', type: 'line', source: 'route', layout: { 'line-cap'
    paint: { 'line-color': '#ffc93c', 'line-width': ['interpolate', ['exponential
});
map.on('error', e => { if (!map.loaded() && !st._mapErr) { st._mapErr = 1; consol
function emptyLine() { return { type: 'Feature', geometry: { type: 'LineString',
function drawRoute(coords) {
  const s = map.getSource('route'); if (s) s.setData({ type: 'Feature', geometry:
}
function updateMeDot(p) {
  if (!meMarker) { const el = document.createElement('div'); el.className = 'me-d
  else meMarker.setLngLat(p);
}
map.on('click', e => {
  if (st.mode !== 'plan') return;
  setDest([e.lngLat.lng, e.lngLat.lat], 'نقطة على الخريطة', true);
});
```

```
['dragstart', 'zoomstart', 'rotatestart', 'pitchstart'].forEach(ev => map.on(ev,
  if (st.mode !== 'nav' || !e.originalEvent) return;
  st.free = true; $('bCenter').classList.remove('hidden');
  clearTimeout(st.freeTimer); st.freeTimer = setTimeout(recenter, 10000);
}));
function recenter() { st.free = false; $('bCenter').classList.add('hidden'); }

/* ---------------- البحث داخل حائل ---------------- */
let qT, qCtl;
$('q').addEventListener('input', () => {
  clearTimeout(qT);
  const v = $('q').value.trim();
  if (v.length < 2) { $('results').innerHTML = ''; return; }
  qT = setTimeout(() => search(v), 650);
});
$('q').addEventListener('keydown', e => { if (e.key === 'Enter') { clearTimeout(q
async function search(q) {
  if (!q) return;
  try {
    if (qCtl) qCtl.abort(); qCtl = new AbortController();
    const url = `${NOM}search?format=jsonv2&limit=8&accept-language=ar&countrycod
    const r = await fetch(url, { signal: qCtl.signal }); const j = await r.json()
    const box = $('results');
    if (!j.length) { box.innerHTML = `<div class="res"><b>ما لقيت نتائج</b><small
    box.innerHTML = '';
    for (const it of j) {
      const d = document.createElement('div'); d.className = 'res';
      const parts = (it.display_name || '').split(/[،,]/).map(s => s.trim());
      d.innerHTML = `<b></b><small></small>`;
      d.querySelector('b').textContent = it.name || parts[0];
      d.querySelector('small').textContent = parts.slice(1, 4).join('، ');
      d.onclick = () => { box.innerHTML = ''; $('q').value = it.name || parts[0];
      box.appendChild(d);
    }
  } catch (e) { if (e.name !== 'AbortError') toast('تعذّر البحث، تأكد من الإنترنت')
}
async function reverseName(p) {
  try {
    const r = await fetch(`${NOM}reverse?format=jsonv2&zoom=17&accept-language=ar
    const j = await r.json(); return j.name || (j.display_name || '').split(/[،,]
  } catch (e) { return null; }
}

/* ---------------- الوجهة والمسار ---------------- */
async function setDest(p, name, rev = false) {
  st.dest = { p, name };
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  if (!destMarker) destMarker = new maplibregl.Marker({ color: '#ff3b3b' }).setLn
  else destMarker.setLngLat(p);
  $('hint').classList.add('hidden'); $('sheet').classList.remove('hidden');
  $('dName').textContent = name; $('dInfo').textContent = 'جاري حساب الطريق…';
  if (rev) reverseName(p).then(n => { if (n && st.dest && st.dest.p === p) { st.d
  const from = st.gps || HAIL;
  if (!st.gps) toast('ما وصلني موقعك بعد — بحسب الطريق من وسط حائل للتجربة');
  try {
    const r = await getRoute(from, p);
    setRoute(r);
    $('dInfo').innerHTML = `المسافة <b>${fmtDist(r.distance)}</b> · الوقت <b>${fm
    const b = new maplibregl.LngLatBounds(); st.coords.forEach(c => b.extend(c));
    map.fitBounds(b, { padding: { top: 90, bottom: 260, left: 40, right: 40 }, du
  } catch (e) { $('dInfo').textContent = 'تعذّر حساب الطريق — جرّب وجهة ثانية'; }
}
async function getRoute(from, to) {
  const url = `${OSRM}${from[0]},${from[1]};${to[0]},${to[1]}?overview=full&geome
  const r = await fetch(url); const j = await r.json();
  if (j.code !== 'Ok' || !j.routes || !j.routes.length) throw new Error(j.code);
  return j.routes[0];
}
function setRoute(r) {
  st.route = r; st.coords = r.geometry.coordinates;
  st.cum = [0];
  for (let i = 1; i < st.coords.length; i++) st.cum.push(st.cum[i - 1] + dist(st.
  st.maxs = (r.legs[0] && r.legs[0].annotation && r.legs[0].annotation.maxspeed)
  st.steps = []; let from = 0;
  for (const leg of r.legs) for (const s of leg.steps) {
    const loc = s.maneuver.location; let bi = from, bd = 1e12;
    for (let i = from; i < st.coords.length; i++) { const d = dist(loc, st.coords
    s.cum = st.cum[bi]; from = bi; st.steps.push(s);
  }
  st.idx = 0; st.ann = {}; drawRoute(st.coords);
}
async function reroute() {
  if (st.rerouting || !st.dest || !st.rawPos) return;
  st.rerouting = true; st.lastReroute = Date.now();
  showAlert('rr', 'جاري تعديل الطريق', 'طلعت عن المسار…', 'ט', 'info', 2500);
  speak('جاري تعديل المسار', false, 'rr', 15000);
  try {
    const r = await getRoute(st.rawPos, st.dest.p);
    setRoute(r); st.fixAlong = 0; st.dispAlong = 0; st.onRoute = true; st.offCoun
  } catch (e) { toast('تعذّر تعديل الطريق'); }
  st.rerouting = false;
}
```

```
/* ---------------- تحديث الموقع ---------------- */
function onFix(p, speedMs, head, t, simAlong) {
  st.rawPos = p;
  const prevKmh = st.kmh;
  st.speed = Math.max(0, speedMs || 0); st.kmh = st.speed * 3.6;
  st.accel = st.accel * 0.6 + (st.kmh - prevKmh) * 0.4;
  if (st.mode !== 'nav') { updateMeDot(p); return; }
  if (simAlong != null) { st.fixAlong = simAlong; st.idx = segAt(simAlong); st.on
  else {
    const s = snap(p);
    if (s && s.d < 35) { st.onRoute = true; st.offCount = 0; st.fixAlong = s.alon
    else { st.onRoute = false; st.offCount++; if (st.offCount >= 3 && Date.now()
  }
  if (head != null && !isNaN(head) && st.speed > 1.5) st.rawHead = head;
  else if (st.lastRaw && dist(st.lastRaw, p) > 4) st.rawHead = bearing(st.lastRaw
  if (!st.lastRaw || dist(st.lastRaw, p) > 4) st.lastRaw = p;
  st.fixT = t;
  setGauge(st.kmh);
  updateNav();
}
function updateNav() {
  if (!st.route) return;
  const along = st.fixAlong, total = st.cum[st.cum.length - 1], rem = Math.max(0,
  const secLeft = st.route.duration * (rem / (st.route.distance || 1));
  $('remD').textContent = fmtDist(rem); $('remT').textContent = fmtTime(secLeft);

  // الطريق الحالي
  let cur = st.steps[0], k = 0;
  for (let i = 0; i < st.steps.length; i++) { if (st.steps[i].cum <= along + 2) c
  $('road').textContent = (cur && (cur.name || cur.ref)) || 'طريق';
  const next = k ? st.steps[k] : st.steps[st.steps.length - 1];
  const dNext = Math.max(0, next.cum - along);
  $('iArrow').innerHTML = arrowSVG(next.maneuver.type, next.maneuver.modifier);
  $('iText').textContent = stepText(next); $('iDist').textContent = `بعد ${fmtDis

  // المسارات (Lanes)
  const lanes = next.intersections && next.intersections[0] && next.intersections
  const lb = $('lanes');
  if (lanes && lanes.length > 1 && dNext < 900) {
    lb.innerHTML = lanes.map(l => {
      const ind = (l.indications && l.indications[0]) || 'straight';
      return `<div class="ln ${l.valid ? 'ok' : ''}">${arrowSVG('turn', ind === '
    }).join('');
  } else lb.innerHTML = '';

  // تعليمات صوتية + تغيير المسار
```

```
  const side = sideOf(next), sideTxt = side < 0 ? 'لليمين' : 'لليسار';
  const key = 's' + k + '_' + Math.round(next.cum);
  if (next.maneuver.type !== 'arrive') {
    if (dNext < 650 && dNext > 250 && !st.ann[key + 'a']) {
      st.ann[key + 'a'] = 1;
      speak(`بعد ${fmtDist(dNext)} ${stepText(next)}${side ? '، غيّر مسارك ' + sid
      if (side) showAlert('lane' + key, `غيّر مسارك ${sideTxt}`, `استعد: ${stepTex
    }
    if (dNext <= 250 && dNext > 60 && !st.ann[key + 'b']) {
      st.ann[key + 'b'] = 1;
      speak(`بعد ${fmtDist(dNext)} ${stepText(next)}`, false, key + 'b', 1);
      if (side) showAlert('lane' + key + 'b', `غيّر مسارك ${sideTxt} الحين`, stepT
    }
    if (dNext <= 40 && !st.ann[key + 'c']) { st.ann[key + 'c'] = 1; speak(`الحين
  }

  // حد السرعة
  const ms = st.maxs[st.idx];
  let lim = null;
  if (ms && ms.speed) lim = ms.unit === 'mph' ? Math.round(ms.speed * 1.609) : Ma
  if (lim !== st.limit) { st.limit = lim; setLimit(lim); }
  const over = lim && st.kmh > lim + 5;
  $('limit').classList.toggle('over', !!over);
  if (over && Date.now() - st.lastOver > 30000) { st.lastOver = Date.now(); speak

  // تقصير الخط الأصفر (يختفي اللي مشيته)
  const now = performance.now();
  if (now - st.lastLineUpd > 1000 && st.onRoute) {
    st.lastLineUpd = now;
    const p = pointAt(along); drawRoute([p, ...st.coords.slice(st.idx + 1)]);
  }

  // الوصول
  if (rem < 25 && !st.arrived) {
    st.arrived = true;
    speak('وصلت إلى وجهتك', true, 'arr', 1);
    showAlert('arr', 'وصلت إلى وجهتك 🎉', st.dest ? st.dest.name : '', '🏁', 'ini
    stopSim(); setTimeout(endNav, 6000);
  }
}

/* ---------------- GPS ---------------- */
function startGPS() {
  if (!navigator.geolocation) { toast('جهازك ما يدعم تحديد الموقع'); return; }
  navigator.geolocation.watchPosition(pos => {
    const p = [pos.coords.longitude, pos.coords.latitude], first = !st.gps;
```

```
    st.gps = p;
    let sp = pos.coords.speed;
    if (sp == null || isNaN(sp)) {
      sp = 0;
      if (st.lastG) { const dt = (pos.timestamp - st.lastG.t) / 1000; if (dt > 0.
    }
    st.lastG = { p, t: pos.timestamp };
    if (first && st.mode === 'plan' && !st.dest) map.flyTo({ center: p, zoom: 15
    if (st.sim) { updateMeDot(p); return; }
    onFix(p, sp, pos.coords.heading, performance.now());
  }, err => {
    if (err.code === 1) toast('اسمح للموقع (GPS) عشان الملاحة تشتغل');
  }, { enableHighAccuracy: true, maximumAge: 500, timeout: 20000 });
}
$('btnMe').onclick = () => { if (st.gps) map.flyTo({ center: st.gps, zoom: 16 });

/* ---------------- محاكاة القيادة (وضع التجربة) ---------------- */
let simAlong = 0, simSpeed = 0;
function startSim() {
  simAlong = 0; simSpeed = 0;
  st.simTimer = setInterval(() => {
    const dt = 0.1, total = st.cum[st.cum.length - 1];
    let k = st.steps.findIndex(s => s.cum > simAlong + 2); if (k < 0) k = st.step
    const dNext = st.steps[k].cum - simAlong, rem = total - simAlong;
    let target = (st.limit || 80);
    if (dNext < 140 && sideOf(st.steps[k])) target = 32;
    if (rem < 120) target = 20;
    if (det.danger) target = Math.min(target, 30);
    simSpeed += clamp(target / 3.6 - simSpeed, -4.5 * dt, 2.6 * dt);
    simAlong = Math.min(total, simAlong + simSpeed * dt);
    onFix(pointAt(simAlong), simSpeed, null, performance.now(), simAlong);
  }, 100);
}
function stopSim() { clearInterval(st.simTimer); st.simTimer = null; }

/* ---------------- بدء وإنهاء الملاحة ---------------- */
async function startNav(sim) {
  if (!st.route) { toast('اختر وجهة أول'); return; }
  if (!sim && !st.gps) { toast('عشان تبدأ القيادة لازم تسمح بالموقع (GPS)'); retur
  if (!sim) { // نعيد حساب الطريق من مكانك الحالي
    try { setRoute(await getRoute(st.gps, st.dest.p)); } catch (e) {}
  }
  // تفعيل الصوت على الجوال (لازم ضغطة)
  try { if ('speechSynthesis' in window) { const u = new SpeechSynthesisUtterance
  st.mode = 'nav'; st.sim = sim; st.arrived = false; st.ann = {}; st.free = false
  st.fixAlong = 0; st.dispAlong = 0; st.idx = 0; st.onRoute = true; st.offCount =
```

```
  st.fixT = performance.now(); st.lastLineUpd = 0;
  st.dispHead = st.coords.length > 1 ? bearing(st.coords[0], pointAt(20)) : 0;
  st.dispPos = st.coords[0];
  document.body.classList.add('nav');
  $('simTag').classList.toggle('hidden', !sim);
  three.show(); onResize();
  try { if ('wakeLock' in navigator) st.wake = await navigator.wakeLock.request('
  speak(sim ? 'بدأ وضع التجربة، توكل على الله' : 'بدأت الملاحة', true, 'start', 1);
  if ($('optCam').checked && !sim) camStart();
  else if (sim) fakeStart();
  if (sim) startSim(); else { setGauge(st.kmh); onFix(st.gps, st.speed, null, per
}
function endNav() {
  stopSim(); camStop(); fakeStop();
  st.mode = 'plan'; st.sim = false;
  document.body.classList.remove('nav', 'danger');
  $('edgeL').classList.remove('on'); $('edgeR').classList.remove('on'); $('alert'
  try { st.wake && st.wake.release(); } catch (e) {} st.wake = null;
  if ('speechSynthesis' in window) speechSynthesis.cancel();
  if (st.route) drawRoute(st.coords);
  map.easeTo({ pitch: 0, bearing: 0, zoom: 14, padding: { top: 0, bottom: 0, left
}
/* ---------------- اختيار سيارتك ولونها ---------------- */
function drawCarPick() {
  document.querySelectorAll('.ct').forEach(b => b.classList.toggle('on', b.datase
  $('cols').innerHTML = CAR_COLORS.map(c => `<button data-c="${c}" class="${c ===
  $('cols').querySelectorAll('button').forEach(b => b.onclick = () => { myCar.col
}
document.querySelectorAll('.ct').forEach(b => b.onclick = () => { myCar.type = b.
drawCarPick();

/* ---------------- عرض القمر الصناعي ---------------- */
let satOn = false;
$('btnSat').onclick = () => {
  satOn = !satOn; $('btnSat').classList.toggle('on', satOn);
  if (!map.getSource('sat')) {
    map.addSource('sat', { type: 'raster', tileSize: 256, maxzoom: 19, attributio
      tiles: ['https://server.arcgisonline.com/ArcGIS/rest/services/World_Imagery
    const firstLine = map.getStyle().layers.find(l => l.type === 'line');
    map.addLayer({ id: 'sat', type: 'raster', source: 'sat', paint: { 'raster-opa
  }
  map.setLayoutProperty('sat', 'visibility', satOn ? 'visible' : 'none');
  toast(satOn ? 'عرض الخريطة' : 'عرض القمر الصناعي');
};

$('btnGo').onclick = () => startNav(false);
```

```
$('btnSim').onclick = () => startNav(true);
$('bEnd').onclick = endNav;
$('bCenter').onclick = recenter;
$('bMute').onclick = () => {
  st.muted = !st.muted; $('bMute').textContent = st.muted ? '🔇 الصوت' : '🔊 سامت'
  if (st.muted && 'speechSynthesis' in window) speechSynthesis.cancel();
};
$('bCam').onclick = () => { if (det.on) { camStop(); if (st.sim) fakeStart(); } e
document.addEventListener('visibilitychange', async () => {
  if (document.visibilityState === 'visible' && st.mode === 'nav' && !st.wake) {
    try { st.wake = await navigator.wakeLock.request('screen'); } catch (e) {}
  }
});

/* ============================================================
   مشهد السيارة ثلاثي الأبعاد (Three.js) — 2016 وهايلكس 2021 موديلات كامري
   ============================================================ */
const three = (() => {
  const T = window.THREE;
  const ok = !!T && typeof CAR_BIN !== 'undefined';
  let renderer, scene, cam, me = null, redL, redR, others = [], shown = false, sT
  const geos = {}, shared = {};

  // فك ضغط الموديلات
  function decode() {
    const s = atob(CAR_BIN), raw = new Uint8Array(s.length);
    for (let i = 0; i < s.length; i++) raw[i] = s.charCodeAt(i);
    const buf = raw.buffer;
    const build = g => {
      const out = {};
      for (const c in g) {
        const m = g[c], q = new Uint16Array(buf, m.vo, m.nv * 3), pos = new Float
        for (let i = 0; i < m.nv; i++) for (let k = 0; k < 3; k++) pos[i * 3 + k]
        const idx = m.big ? new Uint32Array(buf, m.io, m.ni).slice() : new Uint16
        const bg = new T.BufferGeometry();
        bg.setAttribute('position', new T.BufferAttribute(pos, 3));
        bg.setIndex(new T.BufferAttribute(idx, 1));
        bg.computeVertexNormals();
        out[c] = bg;
      }
      return out;
    };
    for (const type in CAR_META) {
      const M = CAR_META[type], w = M.wheel;
      geos[type] = { body: build(M.body), rim: build(M.rim), meta: M };
      // كفر (تاير) مرسوم برمجياً
```

```
    const r = w.r, hw = w.w / 2, pts = [];
    const prof = [[r * 0.72, -hw], [r * 0.93, -hw], [r * 0.99, -hw * 0.8], [r,
    for (const [a, b] of prof) pts.push(new T.Vector2(a, b));
    const tg = new T.LatheGeometry(pts, 28); tg.rotateZ(Math.PI / 2);
    geos[type].tire = tg;
  }
}
function makeEnv() {
  try {
    const pm = new T.PMREMGenerator(renderer);
    const es = new T.Scene();
    const sky = new T.Mesh(new T.SphereGeometry(50, 32, 16), new T.ShaderMateri
      side: T.BackSide, depthWrite: false,
      vertexShader: 'varying vec3 p;void main(){p=position;gl_Position=projecti
      fragmentShader: 'varying vec3 p;void main(){float h=normalize(p).y;vec3 t
    }));
    es.add(sky);
    const lm = new T.MeshBasicMaterial({ color: 0xffffff });
    for (const [x, y, z, s] of [[0, 30, 0, 18], [25, 12, -10, 8], [-25, 12, 10,
    envMap = pm.fromScene(es, 0.02).texture;
  } catch (e) { envMap = null; }
}
function mats() {
  const S = (o) => new T.MeshStandardMaterial(Object.assign({ side: T.DoubleSid
  shared.glass = S({ color: 0x0a0f18, metalness: 0.9, roughness: 0.05, transpar
  shared.clear = S({ color: 0xcfd8e3, metalness: 0.6, roughness: 0.05, transpar
  shared.chrome = S({ color: 0xe3e6ec, metalness: 1, roughness: 0.14 });
  shared.black = S({ color: 0x111215, metalness: 0.3, roughness: 0.55 });
  shared.interior = S({ color: 0x2b2522, metalness: 0, roughness: 0.9 });
  shared.white = S({ color: 0xeeeeee, metalness: 0.1, roughness: 0.5 });
  shared.rim = S({ color: 0xb9bec6, metalness: 0.9, roughness: 0.28 });
  shared.tire = S({ color: 0x141414, metalness: 0, roughness: 0.92 });
  shared.amber = S({ color: 0x6a3a00, emissive: 0xff9a1a, emissiveIntensity: 0.
  shared.light = S({ color: 0xffffff, emissive: 0xfff4dc, emissiveIntensity: 0.
}
function shadowTex() {
  const c = document.createElement('canvas'); c.width = 64; c.height = 128; con
  const gr = g.createRadialGradient(32, 64, 4, 32, 64, 64);
  gr.addColorStop(0, 'rgba(0,0,0,.8)'); gr.addColorStop(0.6, 'rgba(0,0,0,.45)')
  g.fillStyle = gr; g.fillRect(0, 0, 64, 128); return new T.CanvasTexture(c);
}
// يبني سيارة كاملة (كامري أو هايلكس) بلون معيّن
function makeCar(type, color) {
  const G = geos[type], M = G.meta, g = new T.Group();
  const paint = new T.MeshStandardMaterial({ color, metalness: 0.55, roughness:
  const tail = new T.MeshStandardMaterial({ color: 0x5a0000, emissive: 0xff1a1a
```

```
  for (const c in G.body) {
    const m = c === 'paint' ? paint : c === 'tail' ? tail : (shared[c] || share
    g.add(new T.Mesh(G.body[c], m));
  }
  const wheels = [], w = M.wheel;
  for (const [sx, z] of [[-1, w.zf], [1, w.zf], [-1, w.zr], [1, w.zr]]) {
    const wg = new T.Group(), spin = new T.Group();
    for (const c in G.rim) spin.add(new T.Mesh(G.rim[c], c === 'rim' ? shared.r
    spin.add(new T.Mesh(G.tire, shared.tire));
    wg.add(spin); wg.position.set(sx * w.x, w.y, z);
    if (sx > 0) wg.scale.x = -1;
    g.add(wg); wheels.push(spin);
  }
  const [mn, mx] = M.bounds;
  const shd = new T.Mesh(new T.PlaneGeometry((mx[0] - mn[0]) * 1.35, (mx[2] - m
  shd.rotation.x = -Math.PI / 2; shd.position.set(0, 0.02, (mx[2] + mn[2]) / 2)
  g.userData = { tail, paint, wheels, type };
  return g;
}
function init() {
  if (!ok) return;
  const cv = $('three');
  renderer = new T.WebGLRenderer({ canvas: cv, alpha: true, antialias: true });
  renderer.setPixelRatio(Math.min(2, window.devicePixelRatio || 1));
  renderer.setClearColor(0x000000, 0);
  scene = new T.Scene();
  cam = new T.PerspectiveCamera(37, 1, 0.1, 400);
  cam.position.set(0, 4.6, 11.5); cam.lookAt(0, 0, -1.2);          // الخريطة ن
  scene.add(new T.HemisphereLight(0xdfe8ff, 0x22222a, 0.9));
  const dl = new T.DirectionalLight(0xffffff, 1.25); dl.position.set(-6, 12, 8)
  const dl2 = new T.DirectionalLight(0xffd9a0, 0.6); dl2.position.set(6, 5, -10
  sTex = shadowTex();
  try { decode(); } catch (e) { console.warn(e); return; }
  makeEnv(); mats();
  const fade = (() => {
    const c = document.createElement('canvas'); c.width = 4; c.height = 128; co
    const gr = g.createLinearGradient(0, 0, 0, 128);
    gr.addColorStop(0, 'rgba(255,255,255,0)'); gr.addColorStop(0.35, '#fff'); g
    g.fillStyle = gr; g.fillRect(0, 0, 4, 128); return new T.CanvasTexture(c);
  })();
  const mkRed = x => {
    const m = new T.Mesh(new T.PlaneGeometry(3.1, 26), new T.MeshBasicMaterial(
    m.rotation.x = -Math.PI / 2; m.position.set(x, 0.03, -6); scene.add(m); ret
  };
  redL = mkRed(-3.6); redR = mkRed(3.6);
  // سيارات حولك: 4 كامري + 3 هايلكس
```

```
  const plan = ['camry', 'camry', 'camry', 'camry', 'hilux', 'hilux', 'hilux'];
  plan.forEach((t, i) => {
    const c = makeCar(t, CAR_COLORS[(i * 3 + 1) % CAR_COLORS.length]);
    c.visible = false; Object.assign(c.userData, { tid: null, x: 0, z: -40, tx:
  });
  setMine(myCar.type, myCar.color);
}
function setMine(type, color) {
  if (!scene || !geos[type]) return;
  if (me) scene.remove(me);
  me = makeCar(type, color); scene.add(me);
}
function resize(w, h) { if (!renderer) return; renderer.setSize(w, h, false); c
function originY(h) { if (!cam) return h * 0.72; const v = new T.Vector3(0, 0,
let redLt = 0, redRt = 0;
function setSides(l, r) { redLt = l ? 0.55 : 0; redRt = r ? 0.55 : 0; }
function setCars(list) {
  const used = new Set();
  for (const o of list) {
    let c = others.find(x => x.userData.tid === o.id && x.userData.type === o.t
    if (!c) c = others.find(x => x.userData.tid === null && x.userData.type ===
    if (!c) continue;
    if (c.userData.tid !== o.id) {
      c.userData.tid = o.id; c.userData.x = o.x; c.userData.z = o.z - 15;
      c.userData.paint.color.setHex(CAR_COLORS[o.id % CAR_COLORS.length]);
    }
    c.userData.tx = o.x; c.userData.tz = o.z; c.userData.brake = o.brake; used.
  }
  for (const c of others) if (!used.has(c)) { c.userData.tid = null; c.visible
}
function update(dt) {
  if (!renderer || !shown || !me) return;
  const spin = st.speed * dt / 0.32;
  for (const w of me.userData.wheels) w.rotation.x -= spin;
  const braking = st.accel < -0.4 || st.kmh < 2;
  const tl = me.userData.tail;
  tl.emissiveIntensity += ((braking ? 2.8 : 0.6) - tl.emissiveIntensity) * Math
  const pulse = 0.75 + 0.25 * Math.sin(performance.now() / 120);
  redL.material.opacity += (redLt * pulse - redL.material.opacity) * Math.min(1
  redR.material.opacity += (redRt * pulse - redR.material.opacity) * Math.min(1
  for (const c of others) {
    if (c.userData.tid === null) continue;
    c.visible = true;
    c.userData.x += (c.userData.tx - c.userData.x) * Math.min(1, dt * 5);
    c.userData.z += (c.userData.tz - c.userData.z) * Math.min(1, dt * 5);
    c.position.set(c.userData.x, 0, c.userData.z);
```

```
      c.userData.tail.emissiveIntensity = c.userData.brake ? 3 : 0.6;
      for (const w of c.userData.wheels) w.rotation.x -= spin * 0.9;
    }
    renderer.render(scene, cam);
  }
  function show() { shown = true; }
  init();
  return { resize, originY, update, setCars, setSides, show, setMine, ok: () => !
})();

/* ---------------- حلقة الرسم ---------------- */
function onResize() {
  const w = window.innerWidth, h = window.innerHeight;
  three.resize(w, h);
  const y = three.originY(h);
  st.navPad = { top: Math.max(0, 2 * y - h), bottom: 0, left: 0, right: 0 };
}
window.addEventListener('resize', onResize);
let lastFrame = performance.now();
function frame(now) {
  const dt = Math.min(0.1, (now - lastFrame) / 1000); lastFrame = now;
  if (st.mode === 'nav' && st.coords.length) {
    let center, head;
    if (st.onRoute) {
      const pred = st.fixAlong + st.speed * Math.min(1.5, (now - st.fixT) / 1000)
      st.dispAlong += st.speed * dt;
      st.dispAlong += (pred - st.dispAlong) * Math.min(1, dt * 2.5);
      if (Math.abs(pred - st.dispAlong) > 60) st.dispAlong = pred;
      center = pointAt(st.dispAlong);
      const a = pointAt(st.dispAlong - 3), b = pointAt(st.dispAlong + 22);
      head = (a && b && dist(a, b) > 2) ? bearing(a, b) : st.dispHead;
    } else { center = st.rawPos; head = st.rawHead; st.dispAlong = st.fixAlong; }
    st.dispHead = (st.dispHead + angDiff(st.dispHead, head) * Math.min(1, dt * 3)
    if (center) st.dispPos = center;
    if (!st.free && st.dispPos) map.jumpTo({ center: st.dispPos, bearing: st.disp
  }
  three.update(dt);
  requestAnimationFrame(frame);
}

/* ============================================================
   كاميرا الذكاء الاصطناعي (COCO-SSD) — كشف السيارات
   ============================================================ */
const VEH = { car: 1.8, truck: 2.5, bus: 2.6, motorcycle: 0.8 };   // ركبة بالمتر
const det = { on: false, stream: null, model: null, tracks: [], nextId: 1, fakeTi
function loadScript(src) {
```

```
  return new Promise((res, rej) => {
    if (document.querySelector(`script[src="${src}"]`)) return res();
    const s = document.createElement('script'); s.src = src; s.onload = res; s.on
    document.head.appendChild(s);
  });
}
async function camStart() {
  if (det.on) return;
  if (!navigator.mediaDevices || !navigator.mediaDevices.getUserMedia) { toast('ط
  try {
    det.stream = await navigator.mediaDevices.getUserMedia({ video: { facingMode:
  } catch (e) { toast('ما قدرت أشغّل الكاميرا — اسمح لها من الإعدادات'); return; }
  const v = $('vid'); v.srcObject = det.stream;
  try { await v.play(); } catch (e) {}
  det.on = true; $('pip').classList.remove('hidden'); $('bCam').classList.add('on
  if (!det.model && !det.loading) {
    det.loading = true; $('camStat').textContent = 'جاري تحميل الذكاء…';
    try {
      await loadScript(TFJS); await loadScript(COCO);
      det.model = await cocoSsd.load({ base: 'lite_mobilenet_v2' });
      $('camStat').textContent = 'الكشف شغّال ✓';
    } catch (e) { $('camStat').textContent = 'تعذّر تحميل الذكاء'; toast('الإنترنت
    det.loading = false;
  }
  detLoop();
}
function camStop() {
  det.on = false;
  if (det.stream) det.stream.getTracks().forEach(t => t.stop());
  det.stream = null; $('pip').classList.add('hidden'); $('bCam').classList.remove
  clearAnalysis();
}
async function detLoop() {
  if (!det.on) return;
  const t0 = performance.now(), v = $('vid');
  try {
    if (det.model && v.readyState >= 2 && v.videoWidth) {
      const preds = await det.model.detect(v, 12, 0.42);
      process(preds, v.videoWidth, v.videoHeight);
      drawBoxes(preds, v.videoWidth, v.videoHeight);
    }
  } catch (e) {}
  setTimeout(detLoop, Math.max(30, 110 - (performance.now() - t0)));
}
function drawBoxes(preds, W, H) {
  const c = $('boxes'); c.width = W; c.height = H; const g = c.getContext('2d');
```

```
  g.lineWidth = Math.max(2, W / 160); g.font = `${Math.round(W / 22)}px Tajawal`;
  for (const p of preds) {
    if (!VEH[p.class]) continue;
    const [x, y, w, h] = p.bbox;
    g.strokeStyle = '#ffd36b'; g.strokeRect(x, y, w, h);
  }
  g.strokeStyle = 'rgba(255,255,255,.35)'; g.setLineDash([6, 6]);
  g.beginPath(); g.moveTo(W * 0.36, 0); g.lineTo(W * 0.36, H); g.moveTo(W * 0.64,
}
/* تحليل الإطار: مسار كل سيارة (يمين/وسط/يسار) + المسافة + التتبع */
function process(preds, W, H) {
  const now = performance.now() / 1000;
  const fovK = H > W ? 0.89 : 1.2255;            // 2*tan(نصف زاوية الرؤية)
  const objs = [];
  for (const p of preds) {
    if (!VEH[p.class]) continue;
    const [x, y, w, h] = p.bbox, nx = x / W, nw = w / W, nh = h / H, cx = nx + nw
    let edge = (nx < 0.02 || nx + nw > 0.98) && nh > 0.28 && nw < 0.55;
    let lane = cx < 0.36 ? -1 : cx > 0.64 ? 1 : 0;
    if (edge) lane = nx < 0.02 ? -1 : 1;
    if (nw > 0.55) { lane = 0; edge = false; }
    let d = VEH[p.class] / (nw * fovK);
    if (edge) d = Math.min(d, 3);
    const type = p.class === 'truck' || p.class === 'bus' ? 'hilux' : p.class ===
    objs.push({ cx, nx, nw, lane, d, edge, type });
  }
  for (const tr of det.tracks) tr.matched = false;
  for (const o of objs) {
    let best = null, bd = 0.2;
    for (const tr of det.tracks) {
      if (tr.matched) continue;
      const dd = Math.abs(tr.cx - o.cx) + Math.abs(tr.nw - o.nw) * 0.6;
      if (dd < bd) { bd = dd; best = tr; }
    }
    if (!best) { best = { id: det.nextId++, hist: [], d: o.d, cx: o.cx, lane: o.l
    best.prevLane = best.lane;
    best.matched = true; best.miss = 0;
    Object.assign(best, { cx: o.cx, nx: o.nx, nw: o.nw, lane: o.lane, edge: o.edg
    best.votes = best.votes || { camry: 0, hilux: 0, none: 0 }; best.votes[o.type
    best.type = best.votes.hilux > best.votes.camry ? 'hilux' : best.votes.camry
    best.d = best.hist.length ? best.d * 0.55 + o.d * 0.45 : o.d;
    best.hist.push({ t: now, cx: o.cx, d: best.d });
    while (best.hist.length && now - best.hist[0].t > 1.2) best.hist.shift();
  }
  det.tracks = det.tracks.filter(tr => { if (!tr.matched) tr.miss = (tr.miss || 0
  analyze(now);
```

```
}
function analyze(now) {
  let lead = null, L = false, R = false, sideL = false, sideR = false, cut = 0;
  for (const tr of det.tracks) {
    if (tr.miss > 1) continue;
    if (tr.lane === 0 && (!lead || tr.d < lead.d)) lead = tr;
    if (tr.lane === -1 && tr.d < 25) L = true;
    if (tr.lane === 1 && tr.d < 25) R = true;
    if (tr.edge) { if (tr.lane < 0) sideL = true; else sideR = true; }
    // سيارة تدخل مسارك
    if (tr.hist.length >= 3 && tr.d < 35) {
      const h0 = tr.hist[0], h1 = tr.hist[tr.hist.length - 1], dt = h1.t - h0.t;
      if (dt > 0.45) {
        const toward = tr.lane === -1 ? (h1.cx - h0.cx) : tr.lane === 1 ? (h0.cx
        const inner = tr.lane === -1 ? tr.nx + tr.nw > 0.40 : tr.lane === 1 ? tr.
        if (toward / dt > 0.07 && inner) cut = tr.lane;
      }
      if (tr.lane === 0 && tr.prevLane && tr.prevLane !== 0 && now - tr.born > 0.
    }
  }
  // فرملة السيارة الأمامية
  let braking = false;
  if (lead && lead.hist.length >= 3) {
    const h0 = lead.hist[0], h1 = lead.hist[lead.hist.length - 1], dt = h1.t - h0
    if (dt > 0.4 && (h0.d - h1.d) / dt > 2.5 && h1.d < 40) braking = true;
  }
  if (cut) {
    if (showAlert('cut', 'في أحد بيدخل مسارك', cut < 0 ? 'من اليمين — خفّف السرعة'
      { speak('انتبه، في أحد بيدخل مسارك', true, 'cut', 6000); beep(660, 0.15); }
  }
  if (braking && !cut) {
    if (showAlert('brake', 'خفّف السرعة وحافظ على المسافة', 'يارة اللي قدامك تفرمل',
      { speak('السيارة اللي قدامك تفرمل', true, 'brake', 6000); beep(740, 0.15);
  }
  // خطر الصدم: أقل من 10 متر
  const moving = st.sim || st.mode !== 'nav' || st.kmh > 8;
  const danger = !!(lead && lead.d < 10 && moving);
  if (danger !== det.danger) {
    det.danger = danger; document.body.classList.toggle('danger', danger);
    clearInterval(det.dangerBeep);
    if (danger) {
      speak('انتبه! ابعد عن السيارة اللي قدامك', true, 'danger', 5000);
      beep(1000, 0.12, 0.3); det.dangerBeep = setInterval(() => beep(1000, 0.12,
    }
  }
  // نص حالة المسارات
```

```
  const parts = [];
  if (lead && lead.d < 30) parts.push('قدامك سيارة');
  if ((R || sideR) && (L || sideL)) parts.push('في يمينك سيارة ويسارك سيارة');
  else if (R || sideR) parts.push('يمينك في سيارة');
  else if (L || sideL) parts.push('يسارك في سيارة');
  const ln = $('lane');
  ln.textContent = parts.length ? parts.join(' · ') : 'المسار فاضي';
  ln.classList.toggle('busy', parts.length > 0);
  $('lead').innerHTML = lead ? `للأمامية <b>${Math.round(lead.d)} م</b>` : 'للأمامية
  // الخطوط الحمراء الجانبية
  const redLeft = sideL || cut === -1, redRight = sideR || cut === 1;
  $('edgeL').classList.toggle('on', redLeft); $('edgeR').classList.toggle('on', r
  three.setSides(redLeft, redRight);
  // السيارات الافتراضية في المشهد
  const list = [];
  for (const tr of det.tracks) {
    if (tr.miss > 2) continue;
    const dd = clamp(tr.d, 0, 60);
    let x, z;
    if (tr.edge) { x = tr.lane * 3.6; z = -0.6; }
    else { x = clamp((tr.cx - 0.5) / 0.3, -1.25, 1.25) * 3.6; z = -(5 + dd * 0.85
    if (tr.type !== 'none') list.push({ id: tr.id, x, z, brake: tr === lead && br
  }
  three.setCars(list.slice(0, 7));
}
function clearAnalysis() {
  det.tracks = []; det.danger = false; clearInterval(det.dangerBeep);
  document.body.classList.remove('danger');
  $('edgeL').classList.remove('on'); $('edgeR').classList.remove('on');
  $('lane').textContent = 'المسار فاضي'; $('lane').classList.remove('busy');
  $('lead').innerHTML = 'للأمامية <b>--</b>';
  three.setSides(false, false); three.setCars([]);
}

/* ---- سيارات وهمية لوضع التجربة (عشان تشوف كل التنبيهات وانت في البيت) ---- */
function fakeStart() {
  if (det.fakeTimer || det.on) return;
  det.fakeT0 = performance.now();
  det.fakeTimer = setInterval(() => {
    const t = ((performance.now() - det.fakeT0) / 1000) % 36, W = 640, H = 480, o
    const car = (cx, d, cls = 'car') => { const nw = Math.min(0.9, (cls === 'truc
    if (t < 10) car(0.5, 45 - t * 2);                    // قدامك تقرب شوي شوي
    else if (t < 13.5) car(0.5, 25 - (t - 10) * 5.9);    // تنبيه فرملة ثم خطر
    else if (t < 17) car(0.5, 4.4 + (t - 13.5) * 4.5);    // تبتعد
    if (t >= 5 && t < 10) car(0.2, 30, 'truck');                  // يدة يسارك
    if (t >= 18 && t < 24) out.push({ class: 'truck', score: 0.9, bbox: [0.84 * W
```

```
    if (t >= 25 && t < 31) car(0.18 + 0.32 * Math.min(1, (t - 25) / 3), 14, 'truc
    if (t >= 31) car(0.78, 22);                                  // في المسار اليمين
    process(out, W, H);
  }, 120);
}
function fakeStop() { clearInterval(det.fakeTimer); det.fakeTimer = null; clearAn

/* ---------------- تشغيل ---------------- */
buildGauge();
onResize();
startGPS();
requestAnimationFrame(frame);
if (location.protocol === 'file:') toast('ارفع الملفات على GitHub Pages ا يشتغلون
```